

The Universal Decentralized Protocol for Disaster-Resilient Communication

Transforming every VHF radio into a self-organizing emergency node —
enabling any vessel, volunteer or community to join the disaster response grid,
with zero internet, zero infrastructure dependency.

€0

Monthly subscription cost

~\$43

Unit hardware cost

100%

Offline operation

ANY radio

Hardware compatibility

Deep Tech

TRL 5-9

EU Civil Protection

Maritime SAR

Decentralized

Open Protocol

THE CRITICAL GAP

Why Disaster Communication Fails

SCENARIO: Marmara Earthquake — Istanbul, 03:14 AM

GSM towers overloaded and collapsed. Internet fiber lines severed. AIS stations offline.
72 search-and-rescue boats deployed. None can report position, crew capacity or status to command.

CENTRALIZED FRAGILITY

70% of telecom infrastructure fails within 48h of a major disaster. Every SAR system that depends on GSM, satellite or internet becomes instantly blind when infrastructure collapses.

INVISIBLE FIRST RESPONDERS

Civilian fishermen, volunteer divers and local boat owners — who are physically first on scene — have NO way to inject their data into any emergency coordination grid. Their capacity is wasted.

INTEROPERABILITY VOID

Coast Guard, Navy, NGOs and volunteers all use different, incompatible communication systems. Cross-border and multi-agency coordination breaks down at the critical moment.

The result: In the most critical 72 hours of any catastrophe,
the rescuers who are already on scene are invisible, uncoordinated and uncountable.

Current solutions: Satellite terminals (\$1,000-\$2,500 + monthly fees) require subscriptions that fail when most needed.

THE NEREUS SOLUTION

A Decentralized, Radio-Wave Emergency Fabric

Any radio. Any vessel. Any operator. Any disaster. One unified, self-organizing grid.

CORE INNOVATION

Nereus is not a device — it is a protocol layer that transforms any existing VHF radio into an intelligent, self-reporting emergency node. No infrastructure. No subscriptions. No barriers to entry.

ANY NODE	SELF-ORGANIZING	FULLY OFFLINE
DEMOCRATIZED ACCESS	DECENTRALIZED INTELLIGENCE	ZERO INFRASTRUCTURE
- Any VHF radio becomes a Nereus node - Plug-and-play hardware interface (~\$43) - Works with Icom, Kenwood, Yaesu, Motorola - Civilian vessels join as equals to Coast Guard	- Automatic MMSI assignment via MAC+GPS hash - Cluster leader elected by highest MMSI priority - Group of 5 boats broadcast as single aggregated unit - Network self-heals when nodes drop or join	- Bell 202 AFSK over standard VHF channel 70 - GPS sourced from paired smartphone - Power drawn from radio mic port (no battery) 12V vehicle/vessel power or solar fallback

NEREUS PACKET FORMAT: P:4105N2905E | S:20 | HD:330 | E3M2K1 (105N 2905E, Speed: 20, Heading: 330, Inventory: 3 Engineers, 2 Medics, 1 Captain)

DEMOCRATIZATION

From Command & Control to Collective Coordination

The paradigm shift: every radio becomes a sensor, every operator becomes a contributor.

WHO CAN JOIN THE NEREUS GRID

- Professional SAR Units**
Coast Guard, Navy, AFAD
- Civilian Fishing Vessels**
Already on scene, now visible
- Volunteer Organizations**
Delta SAR, NGOs, clubs
- Logistics & Supply Boats**
Fuel, medical resupply
- Shore-Based Volunteers**
Fixed relay stations

HOW DEMOCRATIZATION WORKS

STEP 1

Any vessel with a VHF radio plugs in the Nereus Link (\$43 interface)

STEP 2

Device auto-assigns a unique MMSI number — no registration required

STEP 3

Crew selects role codes: K=Captain, M=Medic, E=Engineer, S=Swimmer, D=Diver

STEP 4

Device broadcasts compact packet every 30 seconds via Bell 202 AFSK

STEP 5

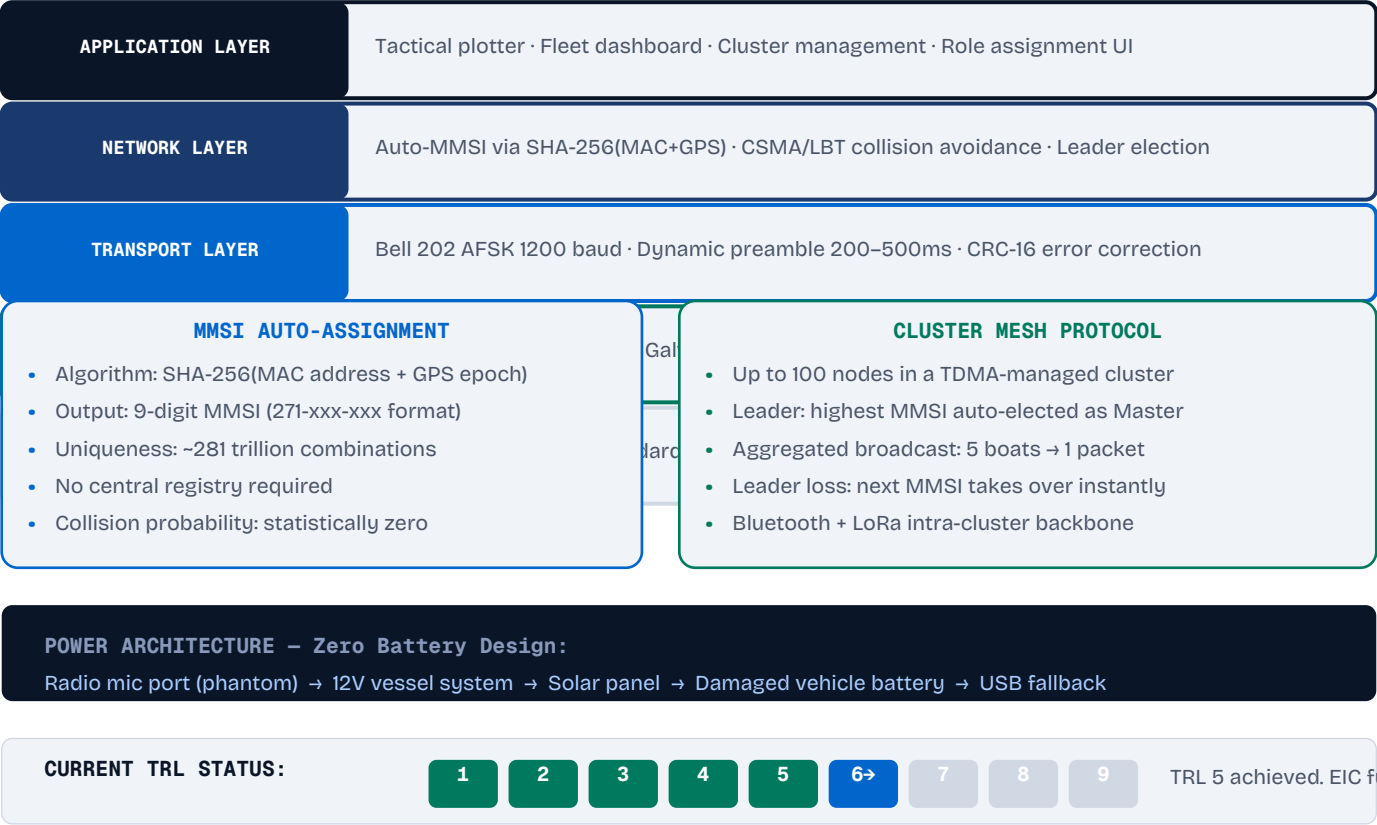
Command center sees ALL units — professional and civilian — on one map

IMPACT: A fishing boat that happened to be near the disaster zone is no longer invisible.
Its crew, position and capabilities are instantly visible to the command center — at zero cost.

System	Entry Cost	Internet Required	Civilian Access	Interoperable
Nereus Protocol	~\$43 (one-time)	NEVER	🔓 ANY operator	🔓 Open standard
Satellite SAR	\$1,000–\$2,500	Always	🔒 Certified only	🔒 Proprietary

TECHNICAL EXCELLENCE

The Deep-Tech Stack Beneath Nereus



MARKET OPPORTUNITY

A Protocol, Not a Product — Capturing Multiple Markets



TARGET VERTICALS



WHY NOW?

EU Directive 2023/1841 mandates critical infrastructure resilience upgrades across member states. The UCPM is actively seeking interoperability with existing systems.

OPEN PROTOCOL STRATEGY: Nereus aims to become the 'HTTP of emergency communication' — licensed to radio manufacturers, embedded in OEM production lines, creating a self-reinforcing network effect.

BUSINESS MODEL

Multi-Channel Revenue from a Single Protocol

B2G GOVERNMENT LICENSING

Grant-funded initial deployments with national civil protection agencies (AFAD-TR, DGSC-EU). Annual protocol license per operational fleet. Target: 20 EU member states within 3 years.

€2M ARR (Year 3)

B2B OEM INTEGRATION

Licensing the Nereus Protocol to radio manufacturers (Icom, Kenwood, Hytera) for factory integration into new VHF hardware production lines. Royalty per unit shipped.

€4M ARR (Year 4)

B2B2G PLOTTER-AS-A-SERVICE

The Nereus Plotter — offline-first tactical command center software — licensed annually to rescue coordination centers, harbor masters and maritime prefectures.

€1.5M ARR (Year 3)

HARDWARE NEREUS LINK UNIT

Direct sales and white-label manufacturing of the \$43 Nereus Link interface box. Target: 50,000 units/year by Year 4 through maritime retail and government procurement.

€3M Revenue (Year 4)

FINANCIAL PROJECTIONS

Year 1	Year 2	Year 3	Year 4	Year 5
€0.3M	€1.2M	€3.5M	€9.8M	€22M

COMPETITIVE LANDSCAPE

Why Nereus Cannot Be Replicated Quickly

SOLUTION	UNIT COST	INTERNET	OFFLINE	OPEN	ANY RADIO	CIVILIAN
Nereus Protocol	~\$43	NEVER	☐	☐	☐	☐
Satellite AIS	\$1,500+	Always	☐	☐	☐	☐
INMARSAT/Iridium	\$2,000+	Always	☐	☐	☐	☐
Standard DSC	Bundled	None	☐	☐	~	☐
LoRa Mesh (other)	\$80–200	None	☐	~	☐	~
PMR446 / CB Radio	\$30–80	None	☐	~	~	☐

THE NEREUS MOAT

Network Effect

Every new node joining increases value for all. The protocol becomes more resilient with scale.

Open Standard Lock-In

By publishing the protocol openly, we create a self-reinforcing ecosystem while retaining commercial leadership.

Regulatory Positioning

First-mover to submit Nereus to ETSI/EN standardization body for EU emergency communications.

Data Sovereignty

100% offline operation means no cloud dependency, no GDPR exposure — a unique EU advantage.

IMPACT & ROADMAP

From Istanbul to a European Standard

30%

Reduction in response time

50%

Operational cost saving

0

Internet dependency

∞

Network value with each node

€0

Monthly subscription

DEVELOPMENT ROADMAP

PHASE 1

NOW

TRL 5

- Prototype complete
- Bell 202 verified
- Cluster logic tested
- UI/UX finalized

PHASE 2

EIC GRANT

TRL 7

- CE/EMC certification
- 100-unit production
- AFAD pilot deployment
- EU partner network

PHASE 3

SCALE

TRL 9

- ETSI standard submission
- OEM partnerships
- 5,000 units deployed
- EU-wide coverage

PHASE 4

GLOBAL STANDARD

- NATO SAR compatibility
- IMO protocol review
- Global licensing
- Open foundation

EU STRATEGIC ALIGNMENT: Horizon Europe Civil Security Cluster 3 · EU UCPM Resilience Mandate · European Critical Infrastructure Directive · Digital Decade 2030 Targets · EU Green Deal (hardware lifecycle extension) · EU Tech Sovereignty Agenda

THE ASK & TEAM

Fund the Protocol That Cannot Fail

EIC ACCELERATOR FUNDING REQUEST

€2.5M

GRANT

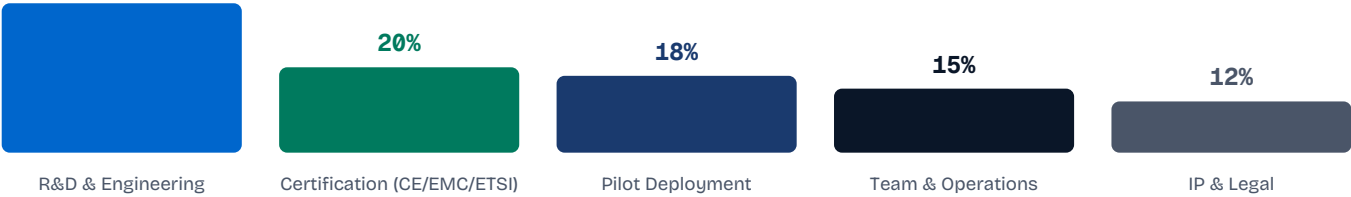
R&D, certification, pilot deployment, team scale-up

€7.5M

EQUITY

Manufacturing scale, EU market entry, OEM partnerships

GRANT ALLOCATION



THE TEAM

Maritime Operations	Systems Architecture	Software & UI/UX	Business & EU Affairs
Ship Captain & SAR Operations	RF & Embedded Systems	Full-Stack & Tactical Systems	Strategy & Grant Management
15+ years maritime command. Delta SAR co-founder. Former marine surveyor. Deep field experience in SAR	Protocol design, Bell 202 implementation, hardware architecture. AFAD technical liaison. Patent inventor (NEREUS system).	Offline-first application development, tactical UI design, API integration and open protocol documentation.	EU funding expertise, maritime industry partnerships, regulatory navigation across 12 EU member states.

"A world where No Signal no longer means No Help."

NEREUS PROTOCOL – The infrastructure for the infrastructure-less.